

PEA515:ANALYTICAL SKILLS-I

L:2 T:2 P:0 Credits:4

Course Outcomes: Through this course students should be able to

CO1 :: apply the procedural fluency with number system and mathematical operations to solve the stated problems.

CO2 :: apply an appropriate approach to solve problems related to percentage and profit & loss.

CO3 :: analyze the reasoning aptitude problems such as blood relation and syllogism.

CO4 :: apply the analytical concepts learnt to solve the questions of ratio & proportion and alligation & mixture.

CO5 :: use the concepts of permutation & combination and probability to solve related problems.

CO6 :: apply the concepts learnt to solve the questions related to direction sense and data interpretation.

Unit I

Number system : classification of numbers, divisibility rules, factorials, unit digit calculation, number of factors, remainder theorem, HCF & LCM

Average : average calculations, inclusion and exclusion, weighted average

Unit II

Percentage : calculation of percentage, percentage comparison, percentage to fraction conversion, successive percent change, problems based on marks, election, population and percentage error

Profit and loss : concept of cost price, selling price and marked price, calculations of profit and loss percentage, problems on discount, faulty weight, number of articles and common selling price

Unit III

Syllogism : logical venn diagrams, ability to relate a certain given group diagrammatically, possibility questions

Blood relation : cracking jumbled up descriptions, relation puzzle, coded relations

Unit IV

Ratio and proportion : concept of ratio and proportion, problems based on ages and partnership

Alligation and mixture : concept based question of alligation and mixtures, replacement based questions

Unit V

Permutation and combination : principles of counting, problems based on selection, arrangement, numbers, words, geometric application, circular arrangement

Probability : concept of probability, classification of events, problems based on coins, dices and cards, conditional probability

Unit VI

Direction sense test : understanding of directions, different types of practice problems

Data interpretation : concept of data interpretation, average and percentage, tabulation, bar graphs, pie charts, line graphs

Text Books:

1. QUANTITATIVE APTITUDE FOR COMPETITIVE EXAMINATIONS by DR. R S AGGARWAL, S Chand Publishing
2. A MODERN APPROACH TO VERBAL & NON-VERBAL REASONING by DR. R S AGGARWAL, S Chand Publishing

References:

1. MAGICAL BOOK ON QUICKER MATHS by M TYRA, BANKING SERVICE CHRONICLE
2. ANALYTICAL REASONING by M.K. PANDEY, BANKING SERVICE CHRONICLE

